order53 RESULTS
PREREG sha256 07e07423…b591 (PREREG.sha256), hashed before tests.
RG45b: PASS, Category A (finite exact checks; data/rg45b.json)
The sources determine the wall data uniquely: order5 §1.4(ii), the order5 completion constants (Z PASS r=6e−16), and the order18 lines() decomposition.
- Lines parallel to c_j^⊥: u1=(0,1) (Q=7/2) and u2=(21,−1) (Q=595/2). T1 checks the orthogonality and T2 checks that each family of lines covers Z^2 exactly once.
- Offsets: θ1 = α_1 and θ2 = α_1+21α_2. Here α is the companion's own shift label from order2 lib2. These are not new symbols.
- Character: f(ν) = e(B(ν,β)) (the lib2 phase_ints phase).
- Multiplicity: m_ℓ = mean of f over one period along line ℓ. It is periodic in ℓ (T3, max dev 0) and does not depend on the base point (T4, 2.2e−16). Tested on all 105 companions × 216 XS.
- The E-arguments are fixed by order5: κ1 = √10·√y and κ2 = √(2y/17).
Limitation: the closed-form A1(R_j) = γ_j·√(π/Q(u_j))·J·Σ_r m_r·B̃1-type sawtooth is not proved here. The Euler–Maclaurin endpoint and sign convention at ℓ+θ ∈ Z (the h(0) value) is not fixed by any written source, and no numeric check of it was preregistered. It stays a sketch.
RG46: SKIPPED. RG45b gives the wall data, not an explicit Ξ_compl or A1−Ξ_compl object. The v12 shadow Ξ is still unidentified with any of these.
Ξ: unidentified. Bridge: NO.
Open next: (a) prereg a numerical check of the A1(R_j) sawtooth formula, with the endpoint convention fixed, against direct Ĥ−H radial asymptotics. (b) Then attempt to match Σ_j A1(R_j) to the v12 Ξ, and run RG46 only if an explicit object results. (c) Phase-aware k≠0 residual bound: not written, left open.
order53 NOTES
- Lock: CORE_FREEZE; new files only in order53/; order5/18/52/51 and papers were read-only (order18 a2.py lines 1–30 exec'd, not modified); MAPS_CANDIDATES untouched; no Ξ/Π/α1 invented (α used only as the existing companion shift label); RG* labels; 35-component law vector; no push or sends.
- Deviations: the first run failed on JSON bool serialization and a sed typo, with no verdict produced. The first draft of the script looped over XS[::6]. Before any verdict, I corrected it to all XS, as PREREG requires.
order53 PREREG — RG45b (and conditional RG46)
Hashed before any test. CORE_FREEZE; new files only; read-only on order5/order18/order52; MAPS_CANDIDATES untouched; no Ξ/Π/α1 invented; 35-component law vector; RG* labels.
RG45b: exact extraction of (m_ℓ, θ, character) for walls (−c1,c2)
Sources: order5 §1.4(ii) (line summation parallel to c_j^⊥), order5 RESULTS completion constants, order18 a2.py lines() (written line decomposition, read-only exec of lines 1–30 only).
Claim to test: lines parallel to c_j^⊥ are u1=(0,1), u2=(21,−1); line label ℓ = B(c_j,ν) ∈ θ_j+Z with θ_1=α_1(companion shift), θ_2=α_1+21α_2; m_ℓ = mean over one period of the companion phase f along the line; character = f = e(B(ν,β)) phase from order2 lib2.
Predefined tests (all exact/integer; PASS iff all true, Category A):
- T1: B_lines(u1,(n,0))-type orthogonality: line direction has B(c1,u1)=0 and B(c2,u2)=0 in the order18 convention (B(c1,v)=v1, B(c2,v)=v1+21v2).
- T2: bijection Z^2→{(ℓ,b)}: v=(n,0)+b u1 and v=(m,0)+b u2 each hit every v in $[-30,30]^2$ exactly once.
- T3: for all 105 companions × all 216 XS: a period P exists on each wall line (order18 period()), and m_ℓ is P_ℓ-periodic in ℓ: |m_{ℓ+N}−m_ℓ| ≤1e−12 with N = lcm of the u-periods over ℓ∈{0..N−1} sample (N := period along the transverse coordinate).
- T4: m_ℓ independent of base point on the line (shift by u): ≤1e−12.
Fail any → document obstruction.
RG46: run iff RG45b yields an explicit Ξ_compl or A1−Ξ_compl object. Otherwise SKIPPED.